UNIVERSITÄTSKLINIKUM
AUGSBURG
II. Medizinische Klinik

Universitätsklinikum Augsburg, Postfach 10 19 20, 86009 Augsburg

Direktor
Prof. Dr. med. Martin Trepel
Sekretariat
Sibylle Blank
Telefon: +49 821 400-2353

Dr. med. Maximilian Schmutz
Stenglinstraße 2, 86156 Augsburg
Telefon: +49 821 400-161030
Fax: +49 821 400-172711
E-Mail: maximilian.schmutz@uk-augsburg.de
Datum:17.12.2025

Prof. Dr. med. Jakob Nikolas Kather
Else Kröner-Fresenius-Center for Digital Health
Technische Universität Dresden
Dresden, Germany

Subject: Application for a Guest Scientist Stay in Your Laboratory

Dear Prof. Kather,

I am writing to apply for a guest scientist stay in your laboratory in Dresden for a defined period of two weeks. I am a physician trained in internal medicine and currently working in hematology/oncology. My clinical and academic interests center on precision oncology and personalized medicine, and I see the responsible implementation of AI as a powerful lever to improve decision-making, efficiency, and patient outcomes.

A recent exchange strongly motivated me to reach out: last week, Prof. Rainer Claus and I met with Marco Gustav and his team during their visit with Prof. Märkl and Nic Reitsam. Your colleagues brought a workstation and models to demonstrate molecular diagnostics based on whole-slide H&E images. Discussing this approach in a real-world setting made the translational potential of computational pathology particularly tangible to us and underscored the value of close collaboration between clinically embedded teams and leading computational groups.

I am part of Prof. Claus' research group, and my work is tightly connected to clinical precision oncology. In addition to my clinical duties, I am clinically responsible for the molecular tumor board (MTB), where we repeatedly face situations in which rapid, cost-effective molecular stratification would have immediate therapeutic consequences. One example we discussed with your team concerns adjuvant decision-making in colorectal cancer, where emerging evidence suggests that patients with alterations in the PI3K pathway may derive a clinically meaningful benefit from aspirin treatment—making reliable identification of eligible patients highly relevant in routine care. In our view, image-based molecular prediction could help reduce diagnostic barriers and accelerate actionable recommendations.

I would like to outline two directions in which a stay in your lab could create strong synergy.

First, we are developing automated, LLM-supported molecular reporting for hematologic neoplasms. Our goal is to support automated drafting of NGS-based reports (approximately 800 reports per year in Augsburg), including variant interpretation. We are building on a corpus of about 4,200 molecular reports and aim to produce clinician-ready drafts that can be reviewed and finalized by the reporting physician. Methodologically, we are exploring a combination of model adaptation and retrieval-based support to handle novel variants while maintaining transparency and traceability (a RAFT-style approach).

Second, we have worked on trial matching using retrieval-augmented generation (unpublished) and performed a head-to-head comparison of real-world MTB cases with a large language model (DOI: 10.1093/oncolo/oyaf293). We also maintain a repository of 1,340 MTB cases spanning 2018 to 2025, providing a strong basis for clinically grounded development and evaluation.

I would be grateful for the opportunity to visit your group for two weeks to present our work, learn from your team's expertise in computational pathology and clinical AI, and define a joint project plan with clear milestones. I am confident that your lab's expertise, paired with our clinical implementation perspective, could lead to productive collaboration with measurable translational and publication impact.

Thank you very much for considering my application. I would welcome the chance to discuss potential directions and practical details in a short call. I am happy to provide my CV, publication list, and a brief project summary at any time.

Sincerely,

Maximilian Schmutz